

Practice Exercise: Single-Agent Inventory & Order Assistant

****Stack:**** C#, `Microsoft.Extensions.AI`, SQL Server

Scenario

You run a small online store. Staff keep asking ad-hoc questions like "which products are almost out of stock?" or "how many orders did customer Rafi place last month?" Right now someone has to open SSMS and write SQL by hand every time.

Build ****one agent, in a loop**** — no sub-agents, no orchestrator — that can explore the schema on its own and answer these questions in plain language.

Why a loop (not a single prompt)

The agent doesn't know the schema up front. It must discover which tables exist, inspect the real column names, then write a query — each step depends on the result of the previous one. If your solution answers in one LLM call without ever inspecting the schema, you haven't built an agent.

What to build

1. Tools
2. Agent
4. Database

Sample questions your agent must handle

- "Which products have fewer than 10 units in stock?"
- "How many orders has customer [name] placed, and what's the total value?"
- "What were total sales in the last 7 days?"
- "List customers who haven't ordered anything in the last 60 days."

Guardrails to keep

- DB login (or app-level check) is ****read-only**** — no writes, no DDL, no batching multiple statements.

- Command timeout on every query.